Miller-Compensated Two-Stage Op-Amp 130nm

Using SKY130 PDK in Xschem

HiuVo

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1. Topology

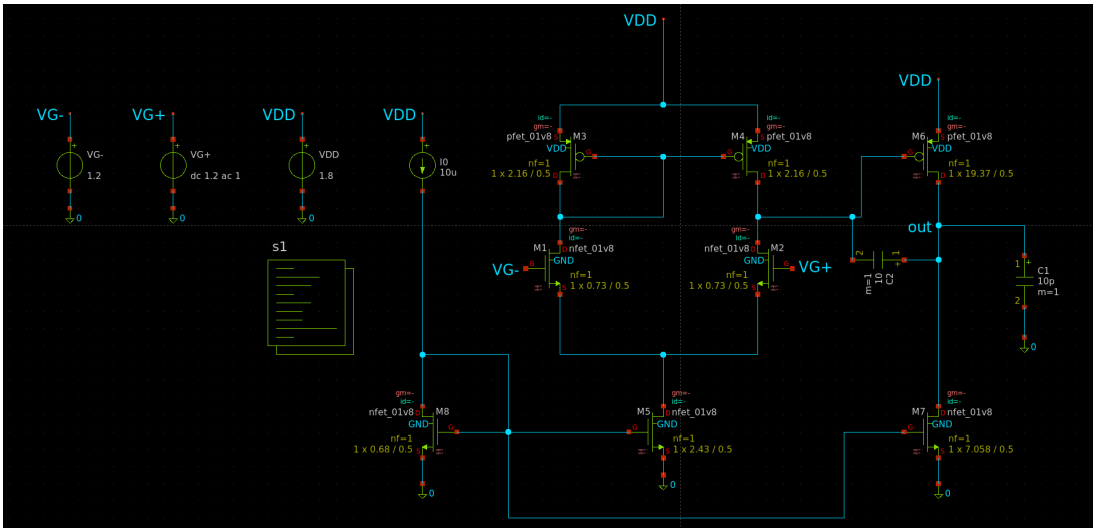


Figure 1: Two Stage Op-Amp Schematic

2. Specifications

- V_{DD}**:** 1.8 V
- **DC Gain:** $\geq$ 60 dB
- **GBW:** 5 MHz
- **Phase Margin:** $\geq 60^{\circ}$
- **Slew Rate:** 10 V$/\mu$sec
- **ICMR:** 0.8 V (min) $\sim$ 1.2 V (max)
- C_L**:** 10 pF
- **Power dissipation:** $\leq 300\mu$W

3. Branch Current Analysis and Calculation

Based on the power dissipation requirement of $\leq 300\mu$W with a supply voltage of $V_{DD} = 1.8$V, the total allowable current for the entire circuit is determined as:

$$I_{\text{total}} = \frac{P_{\text{diss}}}{V_{DD}} \leq \frac{300\mu\text{W}}{1.8\text{V}} \approx 166.7\mu\text{A} \tag{1}$$

- **Tail Current -** I_5**:**

$$I_5 = SR \times C_c = 10 \times 10^6 \times 3 \times 10^{-12} = 30\,\mu\text{A} \tag{2}$$

- **Input Pair - I_1, I_2:**

$$I_1 = I_2 = \frac{I_5}{2} = \frac{30}{2} = 15\,\mu\text{A} \tag{3}$$

- **Active Load - I_3, I_4:** Since they are connected directly in series with M1 and M2, the DC currents flowing through the respective transistors are equal:

$$I_3 = I_1 = 15\,\mu\text{A}, \qquad I_4 = I_2 = 15\,\mu\text{A} \tag{4}$$

- **Bias Reference - I_{ref}, I_8:** Select the reference current $I_{\text{ref}} = 10\,\mu\text{A}$. Since M8 is diode-connected and supplied by the reference current source:

$$I_8 = I_{\text{ref}} = 10\,\mu\text{A} \tag{5}$$

- **Second Stage Bias - I_7:** The minimum current required to charge/discharge the load capacitor C_L is determined from the slew rate requirement:

$$I_{7,\text{min}} \geq SR \times C_L \tag{6}$$

With $SR = 10\,\text{V}/\mu\text{s}$ and $C_L = 10\,\text{pF}$:

$$I_{7,\text{min}} \geq 10 \times 10^6 \times 10 \times 10^{-12} = 100\,\mu\text{A} \tag{7}$$

Therefore, the bias current for the second stage is chosen as:

$$I_7 = 100\,\mu\text{A} \tag{8}$$

- **Second Stage Amplifier - I_6:** M6 (PMOS) and M7 (NMOS) are connected in series in the output branch. In a stable DC state, ignoring leakage and DC load currents:

$$|I_{D6}| = |I_{D7}| \tag{9}$$

Rechecking the total current:

$$I_{\text{total}} = I_8 + I_5 + I_6 = 10 + 30 + 100 = 140\mu\text{A} \leq 166.7\mu\text{A} \text{ (Satisfies the specification)} \tag{10}$$

4. Transistor Sizing using the g_m/I_D Method

To reduce noise and improve matching, the current mirrors need to be deeply biased in the strong inversion region. A low g_m/I_D is selected (around 4 to 8). This design chooses $g_m/I_D \approx 6$. The channel length is fixed at $L = 0.5\mu\text{m}$.

Table 1: Transistor Dimensions (W/L) of the Op-Amp

Transistor	**Type**	**W (μm)**	**L (μm)**
M1, M2	NMOS	0.73	0.5
M3, M4	PMOS	2.16	0.5
M5	NMOS	2.05/ 2.43	0.5
M6	PMOS	19.37	0.5
M7	NMOS	6.83/ 7.058	0.5
M8	NMOS	0.68	0.5

5. DC Analysis

Table 2: Before Adjustment

Transistor	I_D **(μA)**	V_{DS} **(V)**	V_{dsat} **(V)**	**Region**
M1 (NMOS)	12.62	0.308	0.239	Saturation
M2 (NMOS)	12.62	0.308	0.239	Saturation
M3 (PMOS)	12.62	1.233	0.272	Saturation
M4 (PMOS)	12.62	1.233	0.272	Saturation
M5 (NMOS)	25.24	0.257	0.193	Saturation
M6 (PMOS)	96.77	1.510	0.244	Saturation
M7 (NMOS)	96.77	0.289	0.196	Saturation
M8 (NMOS)	10.00	0.859	0.215	Saturation

Table 3: After Adjustment

Transistor	I_D **(μA)**	V_{DS} **(V)**	V_{dsat} **(V)**	**Region**
M1 (NMOS)	14.87	0.305	0.257	Saturation
M2 (NMOS)	14.87	0.305	0.257	Saturation
M3 (PMOS)	14.87	1.259	0.293	Saturation
M4 (PMOS)	14.87	1.259	0.293	Saturation
M5 (NMOS)	29.75	0.234	0.194	Saturation
M6 (PMOS)	111.59	0.931	0.263	Saturation
M7 (NMOS)	111.59	0.868	0.197	Saturation
M8 (NMOS)	10.00	0.859	0.215	Saturation

6. AC Analysis

Table 4: Open-Loop AC Analysis

Parameter	**Uncompensated**	**Compensated** ($C_c = 3$ **pF**)	**Target**
DC Gain	50.57 dB	50.57 dB	≥ 60 dB
GBW	≈ 48 MHz	≈ 3.8 MHz	5 MHz
Phase Margin (PM)	26.8°	66.5°	$\geq 60°$

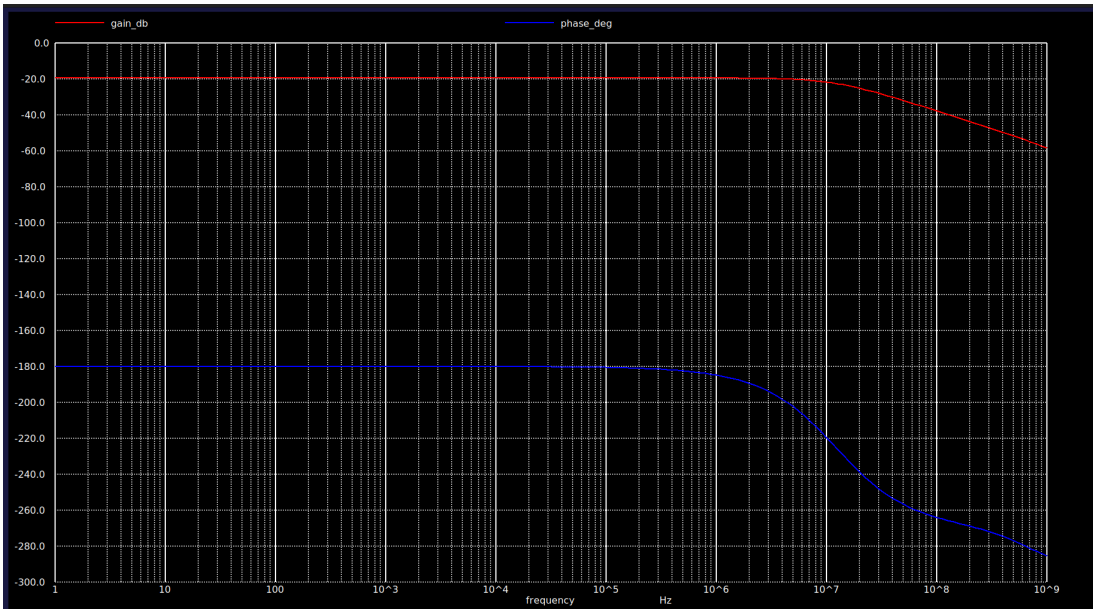


Figure 2: Open-Loop Frequency Response of the Miller-Compensated Op-Amp